Name: _____

Class: X Section: A

Subject: MATHEMATICS Date: _____

Chapter/Topic: COORDINATE GEOMETRY

Delhi Public School
Gurugram I Sector 67A
Nurturing the Whole Child

MULTIPLE CHOICE QUESTIONS

Q1. The distance of the point P(-6, 8) from the origin is

(a) 8 units (b) $2\sqrt{7}$ units

(c) 10 units (d) 6 units

Q2. The distance between the points A(2, -3) and B(2, 2) is

(a) 2 units (b) 4 units

(c) 5 units (d) 3 units

Q3. The distance of the point P(2, 3) from the x-axis is

(a) 2 units (b) 3 units

(c) 1 unit (d) 5 units

Q4. The distance of the point P(-4, 7) from the y-axis is

(a) 4 units (b) -4 units

(c) 7 units (d) $\sqrt{65}$ units

Q5. The coordinates of the midpoint of the line segment joining the points A(-5, 4) and B(7, -8) are

(a) (1, -2) (b) (-1, 2)

(c) (2, -4) (d) (1, 2)

Q6. If the midpoint of the line segment joining A(2a, 4) and B(-2, 3b) is M(1, 2a + 1), then the values of a and b are respectively

(a) a = 2, b = 2 (b) a = 1, b = 2

(c) a = 2, b = 1 (d) a = 1, b = 1

Q7. The point which lies on the perpendicular bisector of the line segment joining the points A(-2, -5) and B(2, 5) is

(a) (0, 0) (b) (0, 2)

(c) (2, 0) (d) (-2, 0)

Q8. The ratio in which the y-axis divides the line segment joining the points P(-4, 2) and Q(8, 3) is

(a) 3 : 1 (b) 1 : 3

(c) 1 : 2 (d) 2 : 1

Q9. The ratio in which the x-axis divides the line segment joining A(2, -3) and B(5, 6) is

(a) 3 : 5 (b) 1 : 2

(c) 2 : 1 (d) 5 : 3

Q10. The point P(1, 2) divides the line segment joining A(-2, 1) and B(7, 4) internally in the ratio

(a) 1 : 2 (b) 2 : 1

(c) 3 : 2 (d) 1 : 3

Q11. If the points A(1, 2), B(0, 0) and C(a, b) are collinear, then

(a) a = b (b) a = 2b

(c) 2a = b (d) a + b = 0

Q12. The perimeter of a triangle with vertices (0, 4), (0, 0) and (3, 0) is

(a) 5 units (b) 11 units

(c) 12 units (d) $7 + \sqrt{5}$ units

Q13. AOBC is a rectangle whose three vertices are A(0, 3), O(0, 0) and B(5, 0). The length of its diagonal is

(a) 5 units (b) 3 units

(c) $\sqrt{34}$ units (d) 4 units

Q14. If the distance between the points (4, p) and (1, 0) is 5 units, then the value of p is

(a) 4 only (b) -4 only

(c) ± 4 (d) 0

Q15. The point on x-axis which is equidistant from A(2, -5) and B(-2, 9) is

(a) (-7, 0) (b) (7, 0)

(c) (0, -7) (d) (-2, 0)

Q16. If three vertices of a parallelogram taken in order are A(-2, 3), B(6, 7) and C(8, 3), then the fourth vertex D is

(a) (0, -1) (b) (0, 1)

(c) (-1, 0) (d) (1, 0)

Q17. The coordinates of the centroid of a triangle with vertices A(-1, 0), B(5, -2) and C(8, 2) are

(a) (12, 0) (b) (6, 0)

(c) (4, 0) (d) (0, 4)

Q18. If the line segment joining the points A(3, 4) and B(k, 6) is divided by the point P(1, 5) in the ratio 1 : 1, then the value of k is

(a) 1 (b) -1

(c) 2 (d) -2

Q19. The area of a triangle with vertices A(3, 0), B(7, 0) and C(8, 4) is

(a) 14 sq. units (b) 28 sq. units

(c) 8 sq. units (d) 6 sq. units

Q20. The point which divides the line segment joining the points (7, -6) and (3, 4) internally in the ratio 1 : 2 lies in

(a) Quadrant I (b) Quadrant II

(c) Quadrant III (d) Quadrant IV

ASSERTION-REASON QUESTIONS

Direction: In the following questions, a statement of assertion (A) is followed by a statement of reason (R). Mark the correct choice as:

(a) Both assertion (A) and reason (R) are true and reason (R) is the correct explanation of assertion (A).

(b) Both assertion (A) and reason (R) are true but reason (R) is not the correct explanation of assertion (A).

(c) Assertion (A) is true but reason (R) is false.

(d) Assertion (A) is false but reason (R) is true.

Q1. Assertion (A): The distance of the point (3, 4) from the origin is 5 units.

Reason (R): The distance of a point P(x, y) from the origin O(0, 0) is given by $OP = \sqrt{(x^2 + y^2)}$.

Q2. Assertion (A): The point (0, 5) lies on the y-axis.

Reason (R): The x-coordinate of any point on the y-axis is zero.

Q3. Assertion (A): The distance between the points A(2, 3) and B(2, 7) is 4 units.

Reason (R): The distance between two points (x_1, y_1) and (x_2, y_2) is given by $\sqrt{((x_2 - x_1)^2 + (y_2 - y_1)^2)}$.

Q4. Assertion (A): The midpoint of the line segment joining the points A(-2, 8) and B(-6, -4) is (-4, 2).

Reason (R): The coordinates of the midpoint of the line joining (x_1, y_1) and (x_2, y_2) are $((x_1 + x_2)/2, (y_1 + y_2)/2)$.

Q5. Assertion (A): The point P(1, 2) divides the line segment joining A(-2, 1) and B(7, 4) in the ratio 1 : 2.

Reason (R): The coordinates of point dividing AB in ratio $m_1 : m_2$ are $((m_1x_2 + m_2x_1)/(m_1 + m_2), (m_1y_2 + m_2y_1)/(m_1 + m_2))$.

Q6. Assertion (A): The line segment joining the points A(3, -1) and B(8, 9) is divided by the x-axis in the ratio 1 : 9.

Reason (R): The ratio in which the x-axis divides the line segment joining (x_1, y_1) and (x_2, y_2) is given by $-y_1 : y_2$.

Q7. Assertion (A): The distance of the point (4, 3) from the y-axis is 3 units.

Reason (R): The distance of any point (x, y) from the y-axis is equal to the absolute value of its x-coordinate.

Q8. Assertion (A): The points A(2, 3), B(4, k) and C(6, -3) are collinear if k = 0.

Reason (R): Three points are collinear if the area of the triangle formed by them is zero.

Q9. Assertion (A): The perimeter of a triangle with vertices (0, 0), (4, 0) and (0, 3) is 12 units.

Reason (R): The area of a right-angled triangle is 1/2 × base × height.

Q10. Assertion (A): If the origin is the midpoint of the line segment joining A(2, 3) and B(x, y), then (x, y) = (-2, -3).

Reason (R): The midpoint of a line segment joining (x_1, y_1) and (x_2, y_2) is $((x_1 + x_2)/2, (y_1 + y_2)/2)$.

Q11. Assertion (A): The points A(-2, 0), B(2, 0) and C(0, 3) form an equilateral triangle.

Reason (R): An equilateral triangle has all three sides of equal length.

Q12. Assertion (A): The y-axis divides the line segment joining P(-4, 2) and Q(8, 3) in the ratio 1 : 2.

Reason (R): The x-coordinate of any point lying on the y-axis is zero.

Q13. Assertion (A): The centroid of the triangle with vertices (3, -5), (-7, 4) and (10, -2) is (2, -1).

Reason (R): The centroid of a triangle divides each median in the ratio 2 : 1.

Q14. Assertion (A): The distance of the point P(a cos θ, a sin θ) from the origin is a units.

Reason (R): For any angle θ, $\cos^2 \theta + \sin^2 \theta = 1$.

Q15. Assertion (A): The point C(1, 2) lies on the line segment joining A(-2, -1) and B(4, 5).

Reason (R): Three points A, B and C are collinear if the slope of AB is equal to the slope of BC.

Q16. Assertion (A): The quadrilateral formed by joining the vertices A(0, -2), B(3, 1), C(0, 4) and D(-3, 1) is a square.

Reason (R): A quadrilateral is a square if all four sides are equal and both diagonals are equal.

Q17. Assertion (A): The point (3, 0) lies on the x-axis.

Reason (R): Every point on the x-axis has its y-coordinate equal to zero.

Q18. Assertion (A): The distance of the point P(-3, -4) from the origin is -5 units.

Reason (R): Distance between any two points is always non-negative.

Q19. Assertion (A): The area of a triangle formed by three collinear points is zero.

Reason (R): Collinear points lie on the same straight line, so no triangle is formed.

Q20. Assertion (A): The coordinates of the point which divides the line segment joining (1, 3) and (2, 7) in the ratio 3 : 4 internally are (11/7, 33/7).

Reason (R): The section formula for internal division is $((m_1x_2 + m_2x_1)/(m_1 + m_2), (m_1y_2 + m_2y_1)/(m_1 + m_2))$.